Hudson

An Extensible Continuous Integration Engine

This article discusses the usefulness of the Hudson CI engine, followed by how to set it up and configure jobs to perform various tasks automatically, or when required.

If you believe in agile software development practices, you already realise the importance of continuous integration. The agile method is to facilitate the delivery of bug-free software to users. The following practices define the truly continuous integration system:

- Members integrate source code frequently
- Each integration is verified by an automatic build
- The code is tested in a mock production environment

The effects of using the continuous integration system are:

- Integration issues are discovered early
- The status of the latest build is easily known
- The current build is available instantly for demonstration or release.

Figure 1 illustrates the architecture of a continuous integration system.

Introducing Hudson

The Hudson system [*http://hudson-ci.org*] is designed precisely to meet continuous integration needs. It is useful to build and test software projects continuously. It provides an easy-to-use interface that makes it easier for developers to integrate changes to a software project. An automated and continuous build increases the productivity of a team.

It is also useful to monitor the externally-executed builds and notify interested parties when something is wrong. Developers can save the changes in the version control system (VCS). The Hudson job(s) monitor the VCS continuously and act accordingly.

Figure 2 shows a screenshot from the Apache build farm that hints how the Hudson home page would look after configuring multiple jobs.

The following are some of the important features of Hudson:

- **Easy installation:** There're no additional installation instructions, nor any need to set up a database separately. Just download the latest stable version of the software (*hudson.war*) from *http://hudson-ci.org* and install it as follows:

```
java -jar hudson.war
```

- **Java and open source:** Hudson is fully written in Java and hence it's platform-independent. It's released under the MIT licence.
- **Easy configuration:** You can